Basic Structures: Sets, Functions, Sequence, Sums, and Matrices

2

Chapter 2

- 2.1 Sets
- 2.2 Set Operations
- 2.3 Functions
- 2.4 Sequence and Summations
- 2.5 Cardinality of Sets
- 2.6 Matrices

2.1 Sets

- Set: the fundamental discrete structure upon which all other discrete structures are built.
- Definition: A set is an unordered collection of objects.
- Set theory is an important branch of mathematics.
  - Many different systems of axioms have been used to develop set theory.
  - Here we are not concerned with a formal set of axioms for set theory. Instead, we will use what is called naïve set theory.

Background [1]

- Geory Cantor (1845-1918): naive set theory
- Paradoxes of naïve set theory:

Bertrand Russell (1872-1970): In a town, the barber shaves all and only those who do not shave themselves. Does the barber shave himself?

Julies Richard (1862-): 每一个整数都可以用若干个字母的词描写出来，描写任何一个数都不止一种方法。现在把整数分为两组，第一组包括所有那些（至少有一种方法）可以用不多于100个字母描写出来的数，第二组包括所有那些不论怎么描写都需要最少是101个字母的数。（第二组中的最小数可以用下面的词表示“the least integer not describable in one hundred or fewer letters.”而该词是小于100的，因此是悖论.）

Background [2]

- Reasons of Paradox： Intuitive notion of an object
- Solution： Clearly define the notion of object, and discriminate class from set by axioms。
- **Ernst Zermelo, Abraham A. Fraenkel, Von Neumann: axioms of set theory**

The concepts of Set [1]

- DEFINITION: The objects in a set are also called the elements, or members, of the set. A set is said to contain it elements.
- Set representation: enumerate (Roster notation), set builder, Venn diagram
- N, Z, R describe natural number, integer, and real number.
- Examples (enumerate):
  - The set *V* of all vowels in the English alphabet can written as *V* = {*a, e, I, o, u*}
  - The set of positive integers less than 100 can be denoted by {1,2,…,99}

The concepts of Set [2]

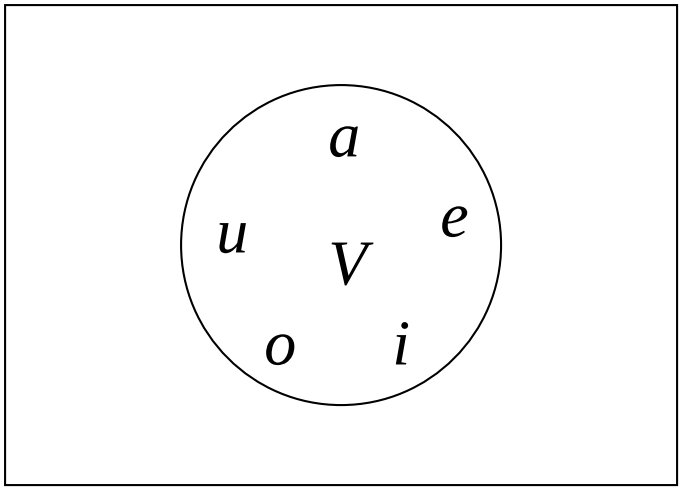
- Examples (set builder):
  - O = { x | x is an odd positive integer less than 10}
  - R = { x | x is a real number}

- DEFINITION: Two sets are equal iff they have the same elements.
- DEFINITION: The set A is said to be a subset of B iff every elements of A is also an element of B. Denoted by $A \subseteq B$, or $\forall x\,(x \in A \rightarrow x \in B)$.
- The null set is a subset of every set $\varnothing \subseteq$ S
- Proper subset: $A \subset B$

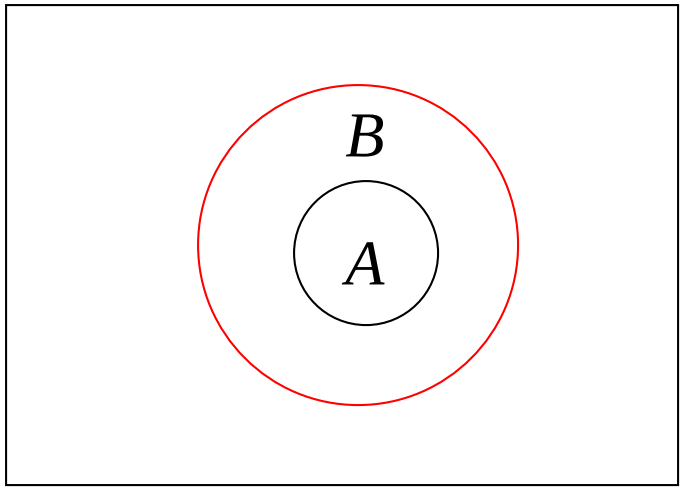
The concepts of Set [3]

- Venn diagram (1881): represents set graphically

Venn diagram for the set of vowels



Venn diagram showing that *A* is a subset of *B*



The concepts of Set [4]

- DEFINITION: Let *S* be a set. If there are exactly *n* distinct elements in *S* where *n* is a nonnegative integer, we say that *S* is a *finite set* and that *n* is the *cardinality* of S. $|S|=n$
- DEFINITION: A set is said to be *infinite* if it is not finite.
- Examples:
  - **Let *A* be the set of odd positive integers less than 10. Then $|A| = 5$.**
  - **The set of positive integers is infinite.**

The Power Set

- DEFINITION: Given a set *S*, the power set of *S* is the set of all subsets of the set *S*. Denoted by *P*(*S*)
- P(S)={ x | x $\subseteq$S }
- | *P*(*S*) |=2^n

- Examples:
  - **The power set of the set {0,1,2}**

    ***P*({0,1,2}) = {∅, {0}, {1}, {2}, {0,1}, {0,2}, {1,2}, {0,1,2}}**
  - **The power set of the empty set**

    ***P*(∅) = {∅}**
  - **The power set of the set {∅}**

    ***P*({∅}) = {∅, {∅}}**

Notes

$$\begin{cases} x \in P(S) \Rightarrow x \subseteq S \\ x \in S \Rightarrow \{x\} \in P(S) \\ S \in P(S) \end{cases}$$

$$P(A) \in P(B) \Rightarrow A \in B$$

Reverse： A=∅, B={∅}√
A={1}, B={{1}} ×

$$A \subseteq B \Rightarrow P(A) \subseteq P(B)$$

Reverse： √

Cartesian Products [1]

- The order of elements in a collection is often important. Since sets are unordered, a different structure is needed to represent ordered collections.
- DEFINITION: The ordered n-tuples $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, …, and a_n as its nth element.
- DEFINITION: Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a,b) \mid a \in A \wedge b \in B\}.$$

Cartesian Products [2]

- DEFINITION: The Cartesian product of the sets A_1, A_2, … , A_n, denoted by

  $A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i, \text{ for } i = 1,2,\ldots,n\}$

- Examples:
  - $A = \{1,2\}$, $B = \{a,b,c\}$

    $A \times B = \{(1,a), (1,b), (1,c), (2,a), (2,b), (2,c)\}$
  - $A = \{0,1\}$, $B = \{1,2\}$, $C = \{0,1,2\}$

    $A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2),$
    $(1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$

Notes

- **If $|A|=m$, $|B|=n$, Then** $|A\times B|=|B\times A|=mn$
- $A\times B\neq B\times A$
- $A\times\phi=\phi\times A=\phi$
- $(x,y)\in A\times B\Rightarrow x\in A \quad and \quad y\in B;(x,y)\notin A\times B\Rightarrow x\notin A \quad or \quad y\notin B$

Using Set Notation with Quantifiers

- $\forall x \in S\ (P(x))$

Truth Sets of Quantifiers

- Given a predicate *P*, and a domain *D*.

  The truth set of *P* to be the set of elements *x* in *D* for which *P*(x) is true. namely,

  - **The truth set of P = {x | p(x) }**

- Problems:

  ∀ *x P*(*x*) is true over the domain *U* ?

  ∃ *x P*(*x*) is true over the domain U ?

Exercises

- P.131: 9, 11, 25, 32, 38, 47

2.2 Set Operations

- Propositional calculus and set theory are both instances of an algebraic system called a

  Boolean Algebra.

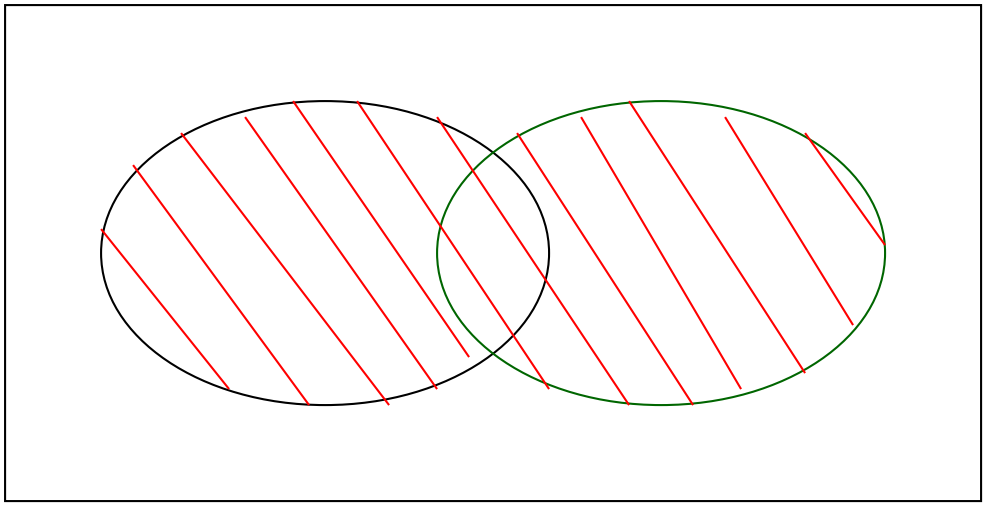
- The operators in set theory are defined in terms of the corresponding operator in propositional calculus.
- As always there must be a universal set *U*. All sets are assumed to be subsets of *U.*

- **Union** ∪
- **Intersection** ∩
- **Difference** -
- **Complement** ~

Union

- Let *A* and *B* be sets. The union of the sets *A* and *B*, denoted by $A \cup B$, is the set that contains the those elements that are either in *A* or in *B*, or in both.

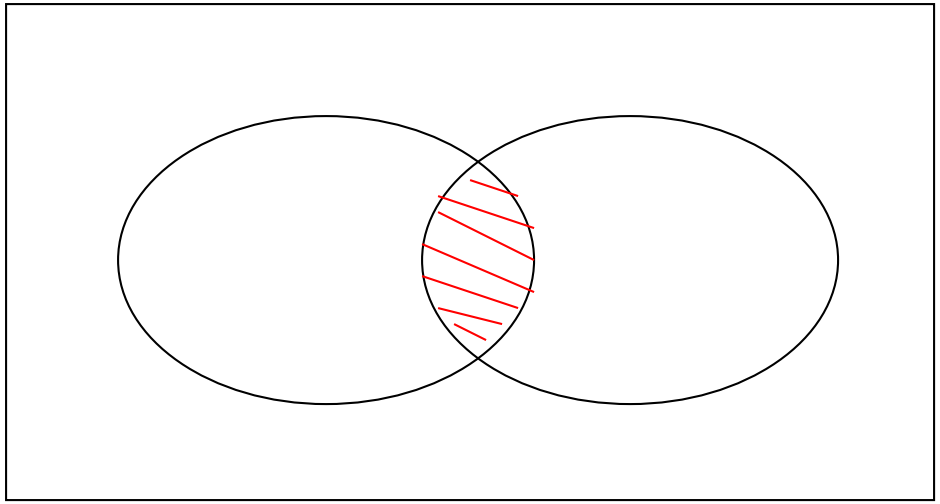
$$\boldsymbol{A \cup B = \{x \mid x \in A \vee x \in B\}}$$



Intersection

- Let *A* and *B* be sets. The intersection of the sets *A* and *B*, denoted by $A \cap B$, is the set containing those elements in both *A* and *B*.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

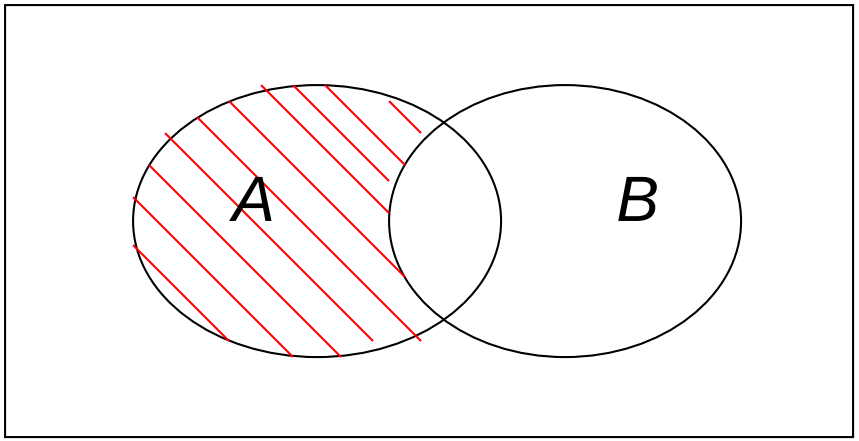


Two sets are called *disjoint* if their intersection is the empty set.

Difference

- Let *A* and *B* be sets. The difference of *A* and *B*, denoted by *A* - *B*, is the set containing those elements that are in *A* but not in *B*. The difference of A and B is also called the complement of B with respect to A.

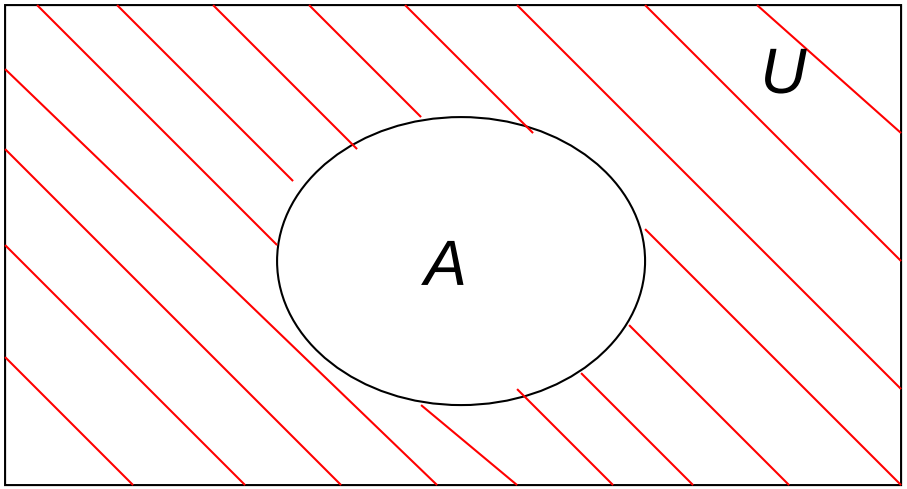
$$A - B = \{x \mid x \in A \wedge x \notin B\}$$



Complement

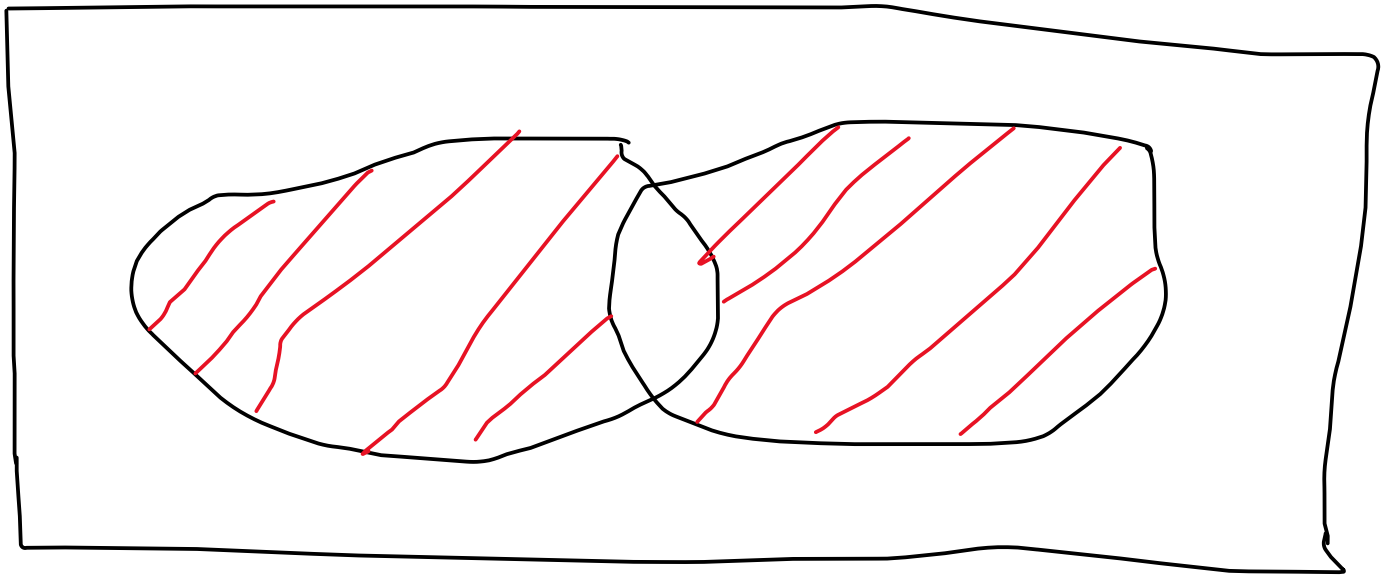
- Let *U* be the universal set. The complement of the set A, denoted by ~A, is the complement of A with respect to *U*. In other words, *U - A.*

$$\sim A = \{x \mid x \notin A\}.$$



Symmetric Difference

$$A \oplus B = (A \bigcup B) - (A \bigcap B)$$



Set Identities [1]

- One way to prove that two sets are equal is to show that one of the sets is a subset of the other and vice versa.
- Example: Prove that $\sim(A \cap B) = \sim A \cup \sim B$ by showing that each set is subset of the other.

1) Suppose that $x \in \sim(A \cap B)$. It follows that $x \notin A \cap B$. This implies that $x \notin A$ or $x \notin B$. Hence, $x \in \sim A$ or $x \in \sim B$. Thus, $x \in \sim A \cup \sim B$. This shows that $\sim(A \cap B) \subseteq \sim A \cup \sim B$

2) Suppose that $x \in \sim A \cup \sim B$. Then $x \in \sim A$ or $x \in \sim B$. It follows that $x \notin A$ or $x \notin B$. Hence $x \notin A \cap B$. This demonstrates that $\sim A \cup \sim B \subseteq \sim(A \cap B)$.

3) Since we have demonstrated that each set is a subset of the other, these two sets must be equal and the identity is proved.

Set Identities [2]

- Another way to verify set identities is to use set builder notation and the rules of logic.
- Example: Use set builder notation and logical equivalences to show $\sim(A \cap B) = \sim A \cup \sim B$

$$\sim(A \cap B) = \{x \mid x \notin A \cap B\}$$
$$= \{x \mid \neg(x \in A \cap B)\}$$
$$= \{x \mid \neg(x \in A \wedge x \in B)\}$$
$$= \{x \mid x \notin A \vee x \notin B\}$$
$$= \{x \mid x \in \sim A \vee x \in \sim B\}$$
$$= \{x \mid x \in (\sim A \cup \sim B)\}$$

Set Identities [3]

Identity	Name
$A \cup \varnothing = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \varnothing = \varnothing$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\sim(\sim A) = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws

Set Identities [4]

Identity	Name
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive laws
$\sim(A \cup B) = \sim A \cap \sim B$ $\sim(A \cap B) = \sim A \cup \sim B$	De Morgan's laws

The set identities given above can be directly used in the other identity proves.

Set Identities [5]

- Additional set identities can be established using those that we have already proved.
- Example: Let A, B, and C be sets. Show that

$$\sim(A \cup (B \cap C)) = (\sim C \cup \sim B) \cap \sim A$$

$$\sim(A \cup (B \cap C)) = \sim A \cap \sim(B \cap C) \quad \text{De Morgan's law}$$
$$= \sim A \cap (\sim B \cup \sim C) \quad \text{De Morgan's law}$$
$$= (\sim B \cup \sim C) \cap \sim A \quad \text{Commutative law}$$
$$= (\sim C \cup \sim B) \cap \sim A \quad \text{Commutative law}$$

Set Identities [6]

- Set identities can be proved using membership tables.
- 1: an element is in a set

  0: an element is not in a set
- Note the similarity between membership tables and truth tables.

- Example: Use a membership table to show that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

(Distributive property)

Set Identities [7]

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Generalized unions and intersections

- DEFINITION: The union of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

$$A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^{n} A_i$$

- DEFINITION: The intersection of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

$$A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^{n} A_i$$

Computer representation of sets [1]

- If to store the elements of the set in an unordered fashion, the operations of computing their union, intersection, difference of two sets would be time-consuming. (Searching time)
- Assume that the universal set U is finite with the cardinality n. First specify an arbitrary ordering of the elements of U, then represent a subset A of U with the bit string of length n, where the ith bit is 1 if a_i belongs to A and is 0 if a_i does not belong to A.
- Union: bitwise *OR*

  Intersection: bitwise *AND*

Computer representation of sets [2]

- Example: What is the bit strings for the set {1,2,3,4,5} and {1,3,5,7,9} with universal set {1,2,3,4,5,6,7,8,9,10}. And Use bit strings to find the union and intersection of these sets.

  {1,2,3,4,5}: 1111100000

  {1,3,5,7,9}: 1010101010

Union: $1111100000 \vee 1010101010 = 1111101010$

{1,2,3,4,5,7,9}

Intersection: $1111100000 \wedge 1010101010 = 1010100000$

{1,3,5}

Exercises

- P.144: 17, 26, 37, 46, 58, 63

2.3 Functions

- Functions
- One-to-one and onto functions
- Inverse functions and compositions of functions
- The graphs of functions
- Some important functions

The concepts of function [1]

DEFINITION:

- Let *A* and *B* be nonempty sets. A *function f* from *A* to *B* is an assignment of exactly one element of *B* to each element of *A*.
- We write $f(a) = b$ if *b* is the unique element of *B* assigned by the function *f* to the element of *A.*
- If *f* is a function from *A* to *B,* we write

$$f : A \rightarrow B$$

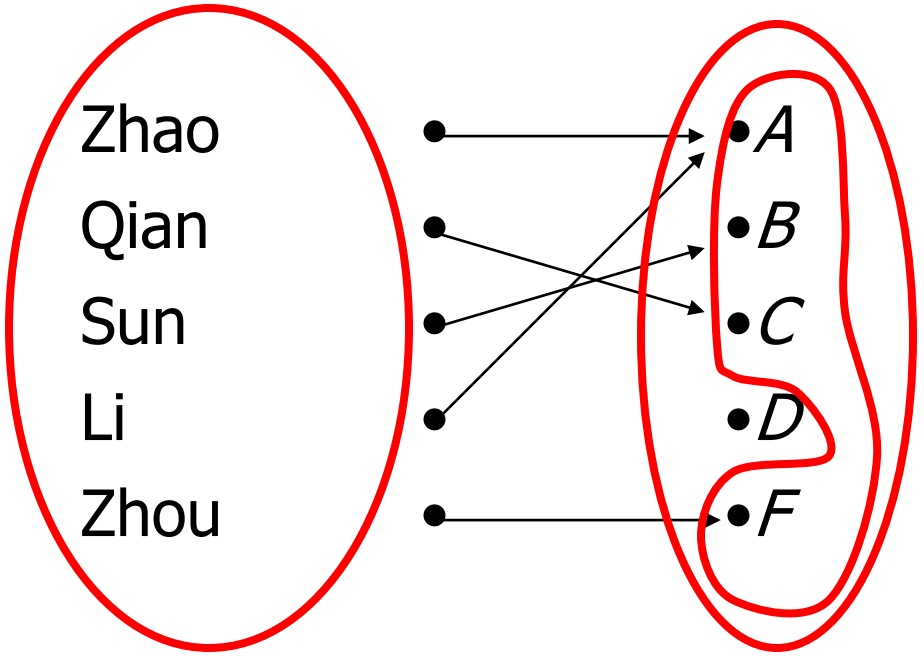
The concepts of function [2]

DEFINITION:

- If *f* is a function from *A* to *B*, we say that *A* is the *domain* （定义域） of *f* and *B* is the （陪域）of *f*.
- If *f(a)* = *b*, we say that *b* is the *image* （像）of *a* and *a* is a *pre-image* （原像）of *b*. The *range*（值域）of *f* is the set of all images of elements of *A*.
- Also, if *f* is a function from *a* to *b*, we say that *f maps* （映射）*A* to *B*. (A given *a* can map to only one *b*)

Example [1]

- Suppose that each student in a class is assigned a letter grade from the set {*A*,*B*,*C*,*D*,*F*}. Let *G* be the function that assigns a grade to a student.



Domain

Codomain

Range

Example [2]

- Consider a function *f* from *Z* to *Z* that assigns the square of an integer to this integer.
- How can you write this function?

$$f(x) = x^2$$

- What is the domain of *f* ?

$$Z$$

- What is the codomain of *f* ?

$$Z$$

- What is the range of *f* ?

$$\{0, 1, 4, 9, \ldots.\}$$

The concepts of function [3]

- DEFINITION: Let f_1 and f_2 be functions from A to **R**. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to **R** defined by

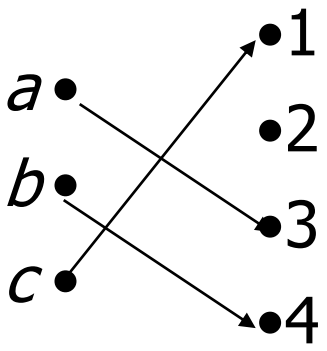
$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$
$$(f_1 f_2)(x) = f_1(x)\, f_2(x)$$

- DEFINITION: Let f be a function from A to B and let S be a subset of A. The image of S is the subset of B that consists of the images of the elements of S. We denote the image of S by $f(S)$, so that

$$f(S) = \{ f(s) \mid s \in S \}$$

One-to-one functions

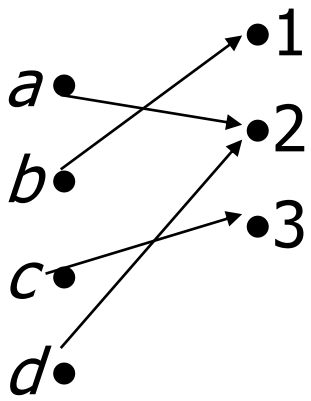
- DEFINITION: A function *f* is said to be *one-to-one*, or *injective*, iff $f(x) = f(y)$ implies that $x = y$ for all x and y in the domain of *f*. A function is said to be an *injection* if it is one-to-one.



- Examples:
  - **$f(x) = x^2$ from the set of integers to the set of integers** X
  - **$f(x) = x+1$** √

Onto functions

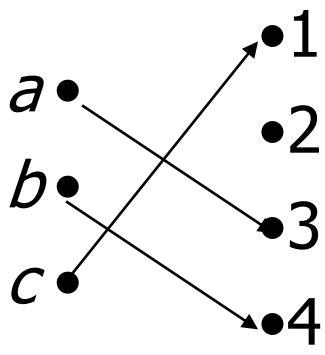
- DEFINITION: A function f from *A* to *B* is called *onto*, or *surjective*, iff for every element $b \in B$ there is an element $a \in A$ with $f(a)=b$. A function *f* is called a *surjection* if it is onto.



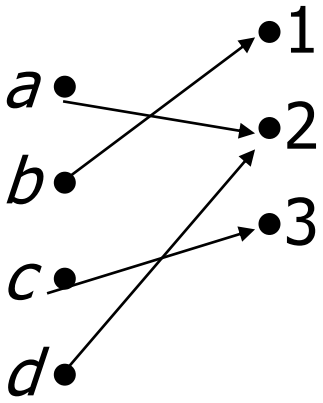
- Examples:
  - ***f*(*x*) = *x*² from the set of integers to the set of integers** X
  - ***f*(*x*) = *x*+1** √

Bijection

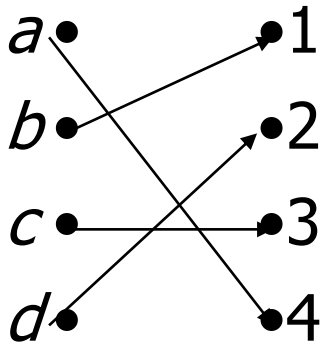
- DEFINITION: The function is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto.



One-to-One



Onto



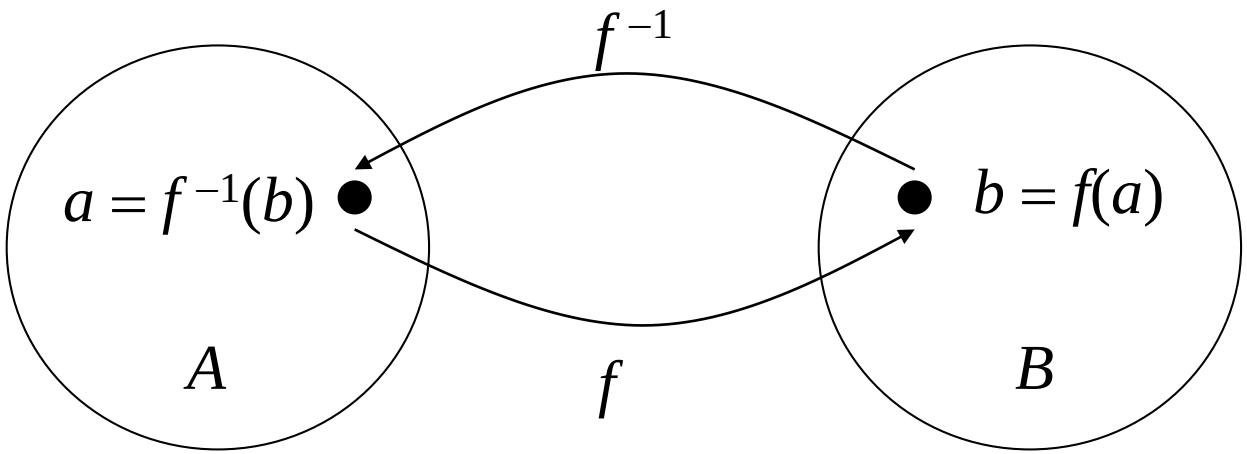
One-to-One Correspondence

Monotonic Functions

- A monotonic function（单调函数）is
  - **either *monotonically* (*strictly*) *increasing***
  - **or *monotonically* (*strictly*) *decreasing***
- Consider a function $f : \mathrm{R} \rightarrow \mathrm{R}$
- *f* is monotonically increasing
  - **if *f*(*x*) < *f*(*y*) whenever *x* < *y***
- *f* is monotonically decreasing
  - **if *f*(*x*) > *f*(*y*) whenever *x* < *y***

Inverse Functions

- DEFINITION: Let *f* be a one-to-one correspondence from *A* to *B*. The *inverse function* of *f* is the function that assigns to an element *b* belonging to *B* the unique element *a* in *A* such that $f(a)=b$. The inverse function of *f* is denoted by f^{-1}. Hence $f^{-1}(b)=a$, when $f(a)=b$.



Examples

- Let *f* be the function from the set of integers to the set of integers such that $f(x) = x+1$. what is its inverse?

  f is one-to-one √

  onto √

  bijection √

  $f^{-1}(y) = y - 1$

- Let *f* be the function from Z to Z with $f(x) = x^2$. Is *f* invertible?

  f is one-to-one X

  No inverse function

Compositions of Functions

- DEFINITION: Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The composition of the functions f and g, denoted by $f \circ g$, is defined by:

$$f \circ g(a) = f(g(a))$$

- $f \circ g$ can't be defined unless the range of g is a subset of the domain of f.
- Example: $f(x) = 2x + 3$, $g(x) = 3x + 2$

$$f \circ g(x) = f(g(x)) = f(3x+2) = 2(3x+2)+3 = 6x+7$$

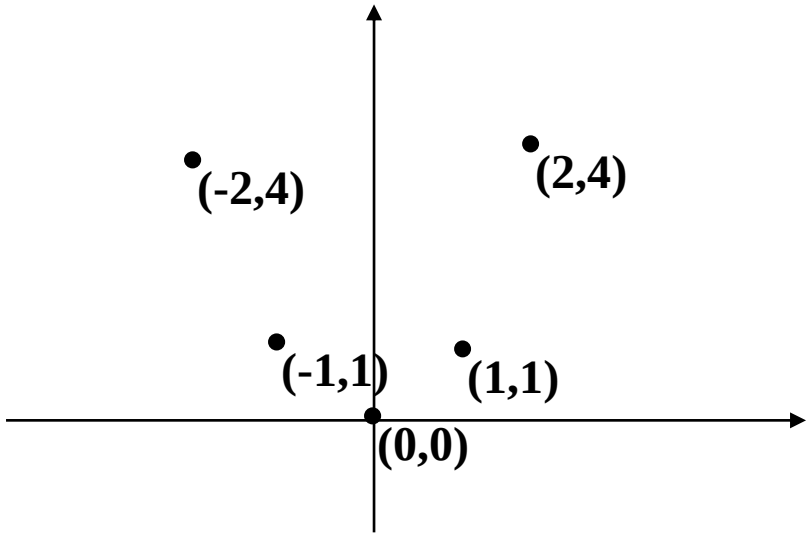
$$g \circ f(x) = g(f(x)) = g(2x+3) = 3(2x+3)+2 = 6x+11$$

The Graphs of Functions

- Let f be a function from the set A to the set B. The *graph* of the function f is the set of ordered pairs

$$\{(a, b) \mid a \in A \text{ and } f(a) = b\}.$$

- Example: Display the graph of the function $f(x) = x^2$ from Z to Z.



Floor

- Let *x* be a real number. The *floor function* is the closest integer less than or equal to *x*.
- Examples:

$$\lfloor \tfrac{1}{2} \rfloor = 0$$

$$\lfloor -\tfrac{1}{2} \rfloor = -1$$

$$\lfloor 3.1 \rfloor = 3$$

$$\lfloor 7 \rfloor = 7$$

Ceiling

- Let x be a real number. The *ceiling function* is the closest integer greater than or equal to x.
- Examples:

$$\lceil \tfrac{1}{2} \rceil = 1$$
$$\lceil -\tfrac{1}{2} \rceil = 0$$
$$\lceil 3.1 \rceil = 4$$
$$\lceil 7 \rceil = 7$$

Floor and Ceiling Functions

TABLE 1 Useful Properties of the Floor and Ceiling Functions.
(n is an integer, x is a real number)

(1a)	$\lfloor x \rfloor = n$ if and only if $n \leq x < n + 1$
(1b)	$\lceil x \rceil = n$ if and only if $n - 1 < x \leq n$
(1c)	$\lfloor x \rfloor = n$ if and only if $x - 1 < n \leq x$
(1d)	$\lceil x \rceil = n$ if and only if $x \leq n < x + 1$
(2)	$x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$
(3a)	$\lfloor -x \rfloor = -\lceil x \rceil$
(3b)	$\lceil -x \rceil = -\lfloor x \rfloor$
(4a)	$\lfloor x + n \rfloor = \lfloor x \rfloor + n$
(4b)	$\lceil x + n \rceil = \lceil x \rceil + n$

Proving Properties of Functions

Example: Prove that x is a real number, then

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$$

Solution: Let $x = n + \varepsilon$, where n is an integer and $0 \le \varepsilon < 1$.

Case 1: $\varepsilon < ½$

- $2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$, since $0 \le 2\varepsilon < 1$.
- $\lfloor x + 1/2 \rfloor = n$, since $x + ½ = n + (1/2 + \varepsilon)$ and $0 \le ½ + \varepsilon < 1$.
- Hence, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + n = 2n$.

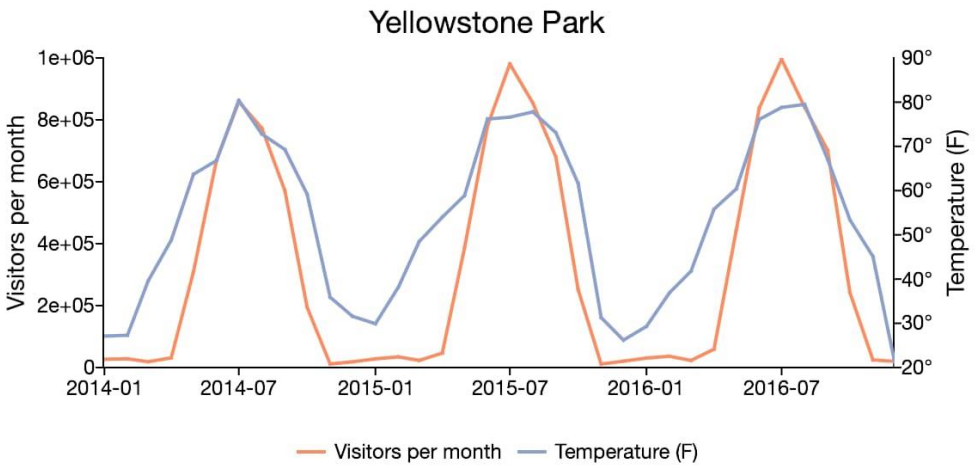
Case 2: $\varepsilon \ge ½$

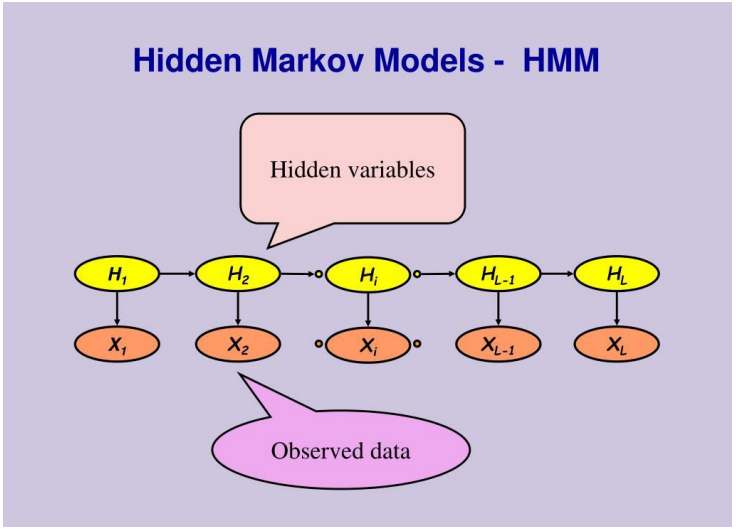
- $2x = 2n + 2\varepsilon = (2n + 1) + (2\varepsilon - 1)$ and $\lfloor 2x \rfloor = 2n + 1$, since $0 \le 2\varepsilon - 1 < 1$.
- $\lfloor x + 1/2 \rfloor = \lfloor n + (1/2 + \varepsilon) \rfloor = \lfloor n + 1 + (\varepsilon - 1/2) \rfloor = n + 1$ since $0 \le \varepsilon - 1/2 < 1$.
- Hence, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + (n + 1) = 2n + 1$.

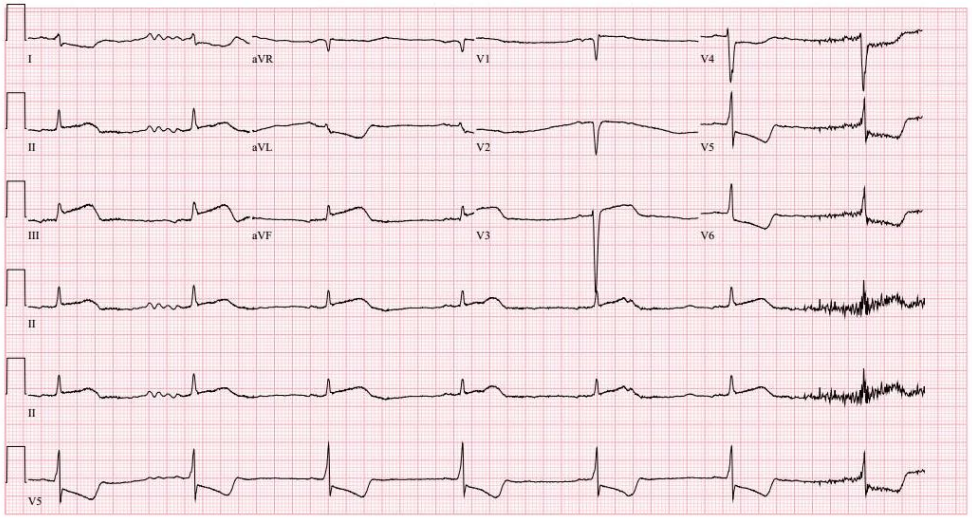
Exercises

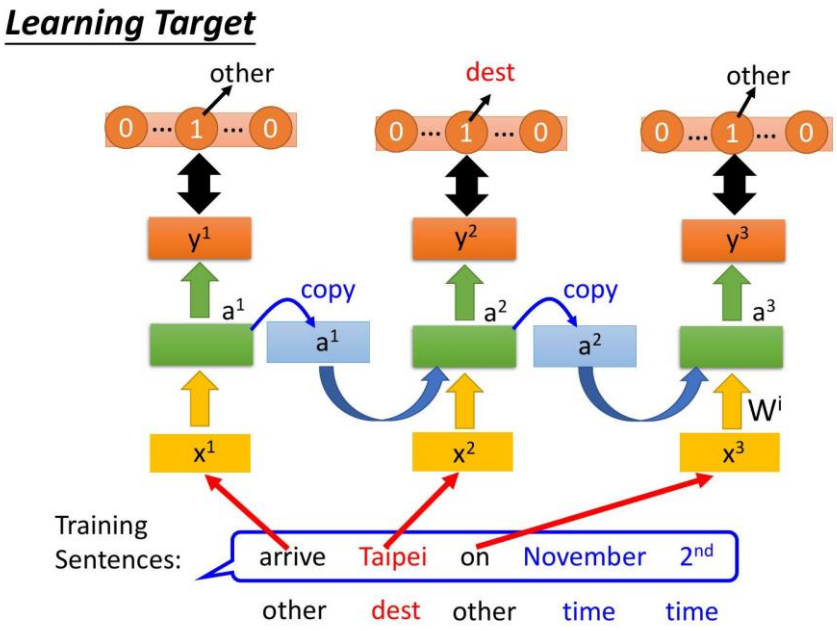
- P.161: 22, 32, 34, 37, 58, 77

2.4 Sequences and summations









Sequences

- DEFINITION: A sequence is a list in which order is taken into account.

  A sequence is a function from a subset of the set of integers (usually either the set {0,1,2,…} or the set {1,2,3,…}) to a set *S*.

- Index of the sequence: a_1, a_2, …, a_n, …

- Examples:

  a, a, b, b, c

  $t_n = n^2-1$, $n \geq 1$ increasing

  $x_n = 1/2^n$ decreasing

Sequences

- Definition: A geometric progression（几何级数）is a sequence of the form

  $$a, ar, ar^2, \ldots, ar^n$$

  where the initial term a and the common ratio r are real number.
- Definition: An arithmetic progression （等差数列）is a sequence of the form

  $$a, a+d, a+2d, \ldots, a+nd$$

  where the initial term a and the common difference d are real number.

Subsequence

- Retain only certain terms of the original sequence, maintaining the order of terms in the given sequence.
- Example:

  2, 4, 8, 16, … , 2^k

  2, 4, 6, 8, 10, 12, … , 2n

Strings

- A string over X is a finite sequence of elements from X.
- The length of a string s is the number of elements in s.
- The empty string, denoted by $\boldsymbol{\lambda}$, is the string that has no term.
- If s and t are two strings, the strings consisting of s followed by t, written st, is called the concatenation of s and t.

Special integer sequences

- A common problem in discrete mathematics is finding a formula or a general rule for constructing the terms of a sequence.
- Deduce a possible formula or rule for the terms of a sequence from the initial terms.
  - **Are there runs of the same value**
  - **Are terms obtained from previous terms by adding the same amount that depends on the position in the sequence**
  - **Are terms obtained from previous terms by multiplying by a particular amount**
  - **Are terms obtained by combining previous terms in a certain way**
  - **Are there cycles among the terms**

Examples

- Find formulas for the sequences with the following first five terms:
  - a) 1, 1/2, 1/4, 1/8, 1/16 $a_n=1/2^n-1$
  - b) 1, 3, 5, 7, 9 $a_n=2n-1$
  - c) 1, -1, 1, -1, 1 $a_n=(-1)^{n+1}$
- How can we produce the terms of a sequence if the first 10terms are 1,2,2,3,3,3,4,4,4,4?
- How can we produce the terms of a sequence if the first 10 terms are 5,11,17,23,29,35,41,47,53,59?

Some useful sequences

- n^2
- n^3
- n^4
- 2^n
- 3^n
- $n!$

Recurrence Relations

- A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses *an* in terms of one or more of the previous terms of the sequence, namely, a_0, a_1,…, a_{n-1}, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to *recursively define* a sequence.)

The *Fibonacci sequence*, $f_0, f_1, f_2, \ldots$, is defined by the initial conditions $f_0 = 0, f_1 = 1$, and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \ldots$.

Summations

$$\Sigma$$

Index of summation, upper/lower limit

Examples

$$\sum_{j=1}^{100} \frac{1}{j}$$

$$\sum_{j=1}^{5} j^2 = 55$$

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij = \sum (i + 2i + 3i) = \sum_{i=1}^{4} 6i = 60$$

$$\sum_{s\in\{0,2,4\}} s = 0 + 2 + 4 = 6$$

Example

- Theorem: If a and r are real numbers and $r \neq 0$, then

$$\sum_{j=0}^{n} ar^j = \begin{cases} \dfrac{ar^{n+1} - a}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1 \end{cases}$$

Some useful summation formula

$$\sum_{k=50}^{100} k^2 = \sum_{k=1}^{100} k^2 - \sum_{k=1}^{49} k^2 = \frac{100\times101\times201}{6} - \frac{49\times50\times99}{6} = 297925$$

$$\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$$

TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1}-a}{r-1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x\rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, \lvert x\rvert < 1$	$\frac{1}{(1-x)^2}$

Some infinite series

$$\sum_{n=0}^{\infty} x^n = \lim_{k\to\infty} \frac{x^{k+1}-1}{x-1} = \frac{-1}{x-1} = \frac{1}{1-x} \qquad \text{if } |x| < 1$$

Exercises

- P.177: 13, 21, 34, 38, 43

2.5 Cardinality of Sets

The Cardinality of a Finite Set

- The cardinality of a finite set was defined to be the number of distinct elements in the set.
- The cardinality of the union of two finite sets:

  The principle of Inclusion-exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| + \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|$$

The Cardinality of an Infinite Set

- How to count the number of elements in an infinite set?
- Are all of the cardinalities of infinite sets same?
- Is the number of positive integers double the number of positive even integers?
- Is the number of real numbers in [a,b] more than the number of real numbers in [0,1]?
- Are there same number of rational numbers and real numbers in [0,1]?

Cardinality [1]

- DEFINITION: The set *A* and *B* have the same cardinality iff there is a one-to-one correspondence from *A* to *B*.

- Example 1: let *A* be the set of 26 lowercase English alphabets. B={1,2,…,26}. Then | *A* | = | *B* |.

Example 2

- Let *A* be the set of real numbers between *a* and *b* (*a*<*b*), and *B* be the set of real numbers between 0 and 1. Show that | *A* | = | *B* |.

Proof:

Let *f* be a function form *A* to *B*.

$$\frac{x-a}{b-a}=\frac{y-0}{1-0}$$

$$y=f(x)=\frac{x-a}{b-a}$$

Then *y* is a bijection from (*a*, *b*) to (0,1).

Hence, | *A* | = | *B* |.

Cardinality [2]

- DEFINITION: A set that is either finite or has the same cardinality as the set of positive integers is called countable. A set is not countable is called uncountable.
- If |A| = |N|, the set A is countable infinite.

- Example: Show that the set of odd positive integers is a countable set.

$$f(n)=2n-1$$

Some Countable Infinite Sets [1]

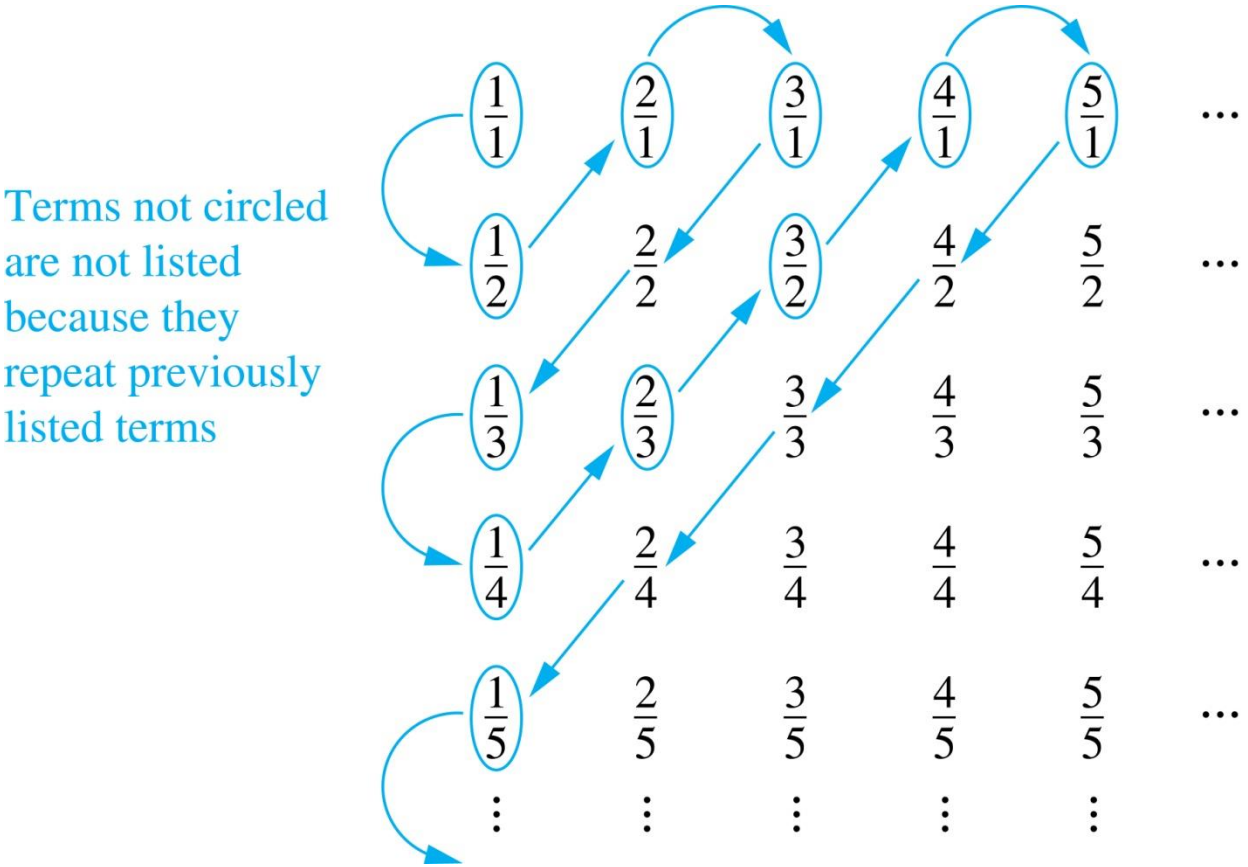
- The set of integers Z

$$Z = \{0,-1,1,-2,+2,\cdots\}$$

$$f(i) = \begin{cases} 2|i| & i < 0 \\ 1 & i = 0 \\ 2i+1 & i > 0 \end{cases}$$

Some Countable Infinite Sets [2]

- The set of positive rational number Q^+ .



The properties of the countable sets

- No infinite set has a smaller cardinality than a countable set.
- The union of two countable sets is countable.
- The union of finite number of countable sets is countable.
- The union of a countable number of countable sets is countable.

Theorem: If A and B are countable sets, then $A \cup B$ is also countable

Proof: Suppose that A and B are both countable sets. Without loss of generality, we can assume that A and B are disjoint. (If they are not, we can replace B by $B - A$, because $A \cap (B - A) = \emptyset$ and $A \cup (B - A) = A \cup B$.) Furthermore, without loss of generality, if one of the two sets is countably infinite and other finite, we can assume that B is the one that is finite.

There are three cases to consider: (i) A and B are both finite, (ii) A is infinite and B is finite, and (iii) A and B are both countably infinite.

Case (i): Note that when A and B are finite, $A \cup B$ is also finite, and therefore, countable.

Case (ii): Because A is countably infinite, its elements can be listed in an infinite sequence $a_1, a_2, a_3, \ldots, a_n, \ldots$ and because B is finite, its terms can be listed as $b_1, b_2, \ldots, b_m$ for some positive integer m. We can list the elements of $A \cup B$ as $b_1, b_2, \ldots, b_m, a_1, a_2, a_3, \ldots, a_n, \ldots$. This means that $A \cup B$ is countably infinite.

Case (iii): Because both A and B are countably infinite, we can list their elements as $a_1, a_2, a_3, \ldots, a_n, \ldots$ and $b_1, b_2, b_3, \ldots, b_n, \ldots$, respectively. By alternating terms of these two sequences we can list the elements of $A \cup B$ in the infinite sequence $a_1, b_1, a_2, b_2, a_3, b_3, \ldots, a_n, b_n, \ldots$. This means $A \cup B$ must be countably infinite.

Proof: Let A and B be countable sets. We will consider four cases.

Suppose both A and B are finite. Then $A \cup B$ is finite, and hence countable.

Suppose one of A and B is finite and the other is countably infinite. Assume without loss of generality that A is finite. Since B is countably infinite, there exists a function $f : B \mapsto \mathbb{Z}^+$ which is a one-to-one correspondence. Say that $f(b_i) = i$, for $b_i \in B$ and $i \in \mathbb{Z}^+$. Let $C = A - B$. Thus $A \cup B = B \cup C$. If $C = \emptyset$, then $A \cup B = B$ which is countably infinite. Thus assume that $C \neq \emptyset$. Suppose that $C = \{c_1, c_2, ...c_n\}$. Let $f' : B \cup C \mapsto \mathbb{Z}^+$ be defined as follows: $f'(c_j) = j$ and $f'(b_i) = n + i$. Clearly, f' is one-to-one and onto. Thus $B \cup C$ is countable. But $B \cup C = A \cup B$, so $A \cup B$ is countable.

Suppose that both A and B are infinite and that $A \cap B = \emptyset$. Given that A and B are both countable, there exists functions $f : \mathbb{Z}^+ \mapsto A$ and $g : \mathbb{Z}^+ \mapsto B$ that are one-to-one correspondences. Consider the function $h : \mathbb{Z}^+ \mapsto A \cup B$ where $h(n) = f(n/2)$ if n is even and $h(n) = g((n + 1)/2)$ if n is odd. Since both f and g are one-to-one, then so is h. Similarly, since f is onto, every element in A is covered and since g is onto, every element in B is coverd, so h is onto. Therefore, h is a one-to-one correspondence, and hence h is countable.

Suppose that both A and B are infinite and that $A \cap B \neq \emptyset$. Let $C = B - A$. Then $A \cup B = A \cup C$ and $A \cap C = \emptyset$. If C is countably infinite, then $A \cup B = A \cup C$ is countable by the previous case. If C is finite, then $A \cup B$ $A \cup C$ is countable byt the second case. In any case, $A \cup B$ is countable.

Lemma 2.1 *Suppose that $S_1, S_2, ... \subset T$ are disjoint countable sets. Then $S = \bigcup_i S_i$ is a countable set.*

Proof: There are bijections $f_i : S_i \to \boldsymbol{N}$ for each i. Let L_k denote the set of elements $s \in S$ such that s lies in some S_i for $i < k$, and $f_i(s) < k$. Note that L_k is a finite set. It has at most k^2 members. The list $L_1, L_2, L_3...$ contains every element of S. Weeding out repeaters, as above, we see that we have listed all the elements of S. Hence S is countable. ♠

The same result holds even if the sets S_i are not disjoint. In the general case, we would define

$$S_k' = S_k - \bigcup_{i=1}^{k-1} S_i,$$

and apply the above argument to the sets $S_1', S_2'....$ The point is that S_i' is countable, the various S' sets are disjoint, and $\bigcup_i S_i = \bigcup_i S_i'$.

SCHRODER-BERNSTEIN THEOREM If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$

- In other words, if there are one-to-one functions f from A to B and g from B to A, then there is a one-to-one
- Because this theorem seems to be quite straightforward, we might expect that it has an easy proof. However, this is not the case because when you have an injection from a set A to a set B that may not be onto, and another injection from B to A that also may not be onto, there is no obvious way to construct a b (bijection from A to B. (Exercise 41)

Example 6: Show that the |(0, 1)| = |(0, 1]|.

Solution: It is not at all obvious how to find a one-to-one correspondence between $(0, 1)$ and $(0, 1]$ to show that $|(0, 1)| = |(0, 1]|$. Fortunately, we can use the Schröder-Bernstein theorem instead. Finding a one-to-one function from $(0, 1)$ to $(0, 1]$ is simple. Because $(0, 1) \subset (0, 1]$, $f(x) = x$ is a one-to-one function from $(0, 1)$ to $(0, 1]$. Finding a one-to-one function from $(0, 1]$ to $(0, 1)$ is also not difficult. The function $g(x) = x/2$ is clearly one-to-one and maps $(0, 1]$ to $(0, 1/2] \subset (0, 1)$. As we have found one-to-one functions from $(0, 1)$ to $(0, 1]$ and from $(0, 1]$ to $(0, 1)$, the Schröder-Bernstein theorem tells us that $|(0, 1)| = |(0, 1]|$. ◀

Uncountable Infinite Sets

- Show that the set of real numbers is an uncountable set.

Solution: contradiction (0-1 is countable)

$$
\begin{aligned}
r_1 &= 0.d_{11}d_{12}d_{13}d_{14}\cdots \\
r_2 &= 0.d_{21}d_{22}d_{23}d_{24}\cdots \\
r_3 &= 0.d_{31}d_{32}d_{33}d_{34}\cdots \\
r_4 &= 0.d_{41}d_{42}d_{43}d_{44}\cdots \\
&\vdots
\end{aligned}
$$

Cantor diagonalization argument

New real number

$$
d_i = \begin{cases} 4 & \text{if } d_{ii} \neq 4 \\ 5 & \text{if } d_{ii} = 4 \end{cases}
$$

Example

- It is said to have the cardinality of the continuum, c.
- The set of real numbers R=(-∞,+∞) has the same cardinality as the set (0,1) .

$$|R| = |(0,1)| = \aleph$$

Proof:

Let $f(x)=\mathrm{tg}(x)$.

$$\because \left|(-\frac{\pi}{2},\frac{\pi}{2})\right| = |(0,1)| \quad \therefore |R| = |(0,1)|$$

Computability

We say that a function is computable if there is a computer program in some programming language that finds the values of this function. If a function is not computable we say it is uncomputable.

Computability

Show that the set of finite strings S over a finite alphabet A is countably infinite.

Proof: Assume an alphabetical ordering of symbols in A. Show that the strings can be listed in a sequence. First list

1. All the strings of length 0 in alphabetical order.
2. Then all the strings of length 1 in lexicographic (as in a dictionary) order.
3. Then all the strings of length 2 in lexicographic order.
4. And so on.

This implies a bijection from N to S and hence it is a countably infinite set.

Computability

Show that the set of functions from the set of positive integers to itself is uncountable.

Assume all function f_1, f_2, f_3…, can be listed in a sequence.
Proof by contradiction and Cantor's diagonal argument.

Construct a function $g(n) = f_n(n) + 1$
By our initial assumption, g must appear in the enumeration f_1, f_2, f_3…
Let $g=f_m$ for some positive integer m. By definition of the enumeration, for n=m: $f_m(m)=g(m)$

This is a **contradiction**.

The properties of the uncountable sets

- Theorem: The cardinality of the power set of an arbitrary set has a greater cardinality than the original arbitrary set.

- The Continuum Hypothesis: The continuum hypothesis (CH) asserts that there is no cardinal number *a* such that $\aleph_n < a < \aleph_{n+1}$.

The Continuum Hypothesis

- Cantor formulated CH, and spent the last years of his career unsuccessfully trying to prove it. His failure obsessed him and caused recurring bouts of serious depression.
- To prove or disprove CH was the first problem on David Hilbert's famous 1900 list of important unsolved problems in mathematics.
- Similarly, attempts to prove or disprove CH, or to prove it undecidable, consumed most of Kurt Godel's 36 years at the Institute for Advanced Study. But he did produce an important partial proof. In 1938, Godel showed that CH cannot be disproved from the axioms of Zermelo-Fraenkel (ZF) set theory.
- In 1963 Paul Cohen showed that CH cannot be proved from the ZF axioms. Together,Godel and Cohen's results show that CH is independent of the ZF axioms: neither it nor its negation can be derived from them. Among other things, this means that CH is undecidable in ZF. After Euclid's parallel postulate, CH was the first major conjecture to be proved undecidable by standard mathematics. Its independence also means that CH is just as consistent with standard set theory as CH itself. This has allowed set theorists to develop set theories with and without it, just as geometers can develop geometries with or without Euclid's parallel postulate. Set theory without CH rather than CH is usually called Non-Cantorian set theory.

整数的幂集势等于实数的势

- 首先,tan(π(2x-1)/2)给出了(0,1)到R的双射,因此二者基数相等.
  只需证明自然数集N的幂集P(N)与(0,1)基数相等.
  直接构造双射比较困难,所以借助Cantor-Bernstein定理:
  若两个集合分别有到对方的单射,则二者基数相等.
  构造f: P(N) → (0,1).
  若S为N的非空子集,定义f(S) = ∑{n ∈ S} 1/10^(n+1).
  当n ∈ S,则f(S)的10进制小数小数点后第n+1位为1,否则为0.
  另外补充定义f(∅) = 0.2.
  构造g:(0,1) → P(N).
  设a ∈ (0,1),定义g(a) = {[(1+a)10^k] | k ∈ N} ([x]表示不超过x的最大整数).
  例如对a = π-3.14 = 0.001592653...,
  有g(a) = {1,10,100,1001,10015,100159,1001592,...}.
  即g(a)是由1+a截断到不同位数得到的,加1则是为了在前面补0.
  不难验证f,g都是单射,故P(N)与(0,1)的基数相等,进而与R的基数相等.

The cardinality of the power set of an arbitrary set has a greater cardinality than the original arbitrary set.

定理：对任意集合 A，有 $|A|<|P(A)|$。
步骤 1：先证明 $|A|\leq|P(A)|$
定义函数 $f:A\to P(A)$，使得对任意 $a\in A$，$f(a)=\{a\}$。
验证单射：若 $f(a1)=f(a2)$，则 $\{a1\}=\{a2\}$，根据集合相等的定义，必有 $a1=a2$。
因此 f 是单射，根据基数大小的定义，$|A|\leq|P(A)|$。
步骤 2：用反证法证明 $|A|\neq|P(A)|$
假设存在**双射函数** $g:A\to P(A)$（即 $|A|=|P(A)|$），我们构造一个矛盾集合：
$B=\{x\in A \text{ and } x\notin g(x)\}$
由于 B 是 A 的子集（$B\subseteq A$），因此 $B\in P(A)$。
因为 g 是双射，所以**存在唯一的** $y\in A$，使得 $g(y)=B$。
现在我们分析 y 是否属于 B，会出现两种矛盾情况：
若 $y\in B$：根据 B 的定义，$y\in/g(y)$，但 $g(y)=B$，因此 $y\notin B$ —— 矛盾。
若 $y\notin B$：根据 B 的定义，$y\in g(y)$，但 $g(y)=B$，因此 $y\in B$ —— 矛盾。

This theorem was proved by Cantor in 1891 and led him in 1895 to what is now known as Cantor's paradox: if U is the universe of all sets, then every subset of U, being a set, is an element of $\wp(U)$. Thus, $\wp(U)\subset U$, so (it would seem) the cardinality of $\wp(U)$ cannot be greater than the cardinality of U. But this contradicts Cantor's theorem. (For an interesting discussion of why Cantor did not see this and other set-theoretic paradoxes as negative (until Russell's), but as positive results, see Dauben [1, pp. 242-247].) This paradox was not published until 1932, but word of its discovery spread and reached Russell in 1901, whereupon he constructed his paradox. Russell's paradox can be seen as the ultimate, set-theoretic application of Cantor's diagonal method: diagonalize out of the *Universe*!

Exercises

- P.186: 21, 28, 29, 38, 39, 40, 41

2.6 Matrices

- **Matrix Arithmetic**
- **Transposes and Powers of Matrices**
- **Zero–One Matrices**

$$A \wedge B$$

$$A \vee B$$

$$A \bullet B$$

Exercises

- P.193: 19, 20, 27, 29, 34